

Reading the Moon

A Public Science Platform and Curriculum

The Moon is my beautiful impossibility.

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Program Vision

Reading the Moon: A Public Science Platform and Curriculum is built on the premise that interested members of the public can participate in the frontier of planetary science and AI, not merely observe it. The platform connects two strands:

1. **Strand A: Using AI to investigate the Moon**
2. **Strand B: Building and understanding AI with lunar data**

Strand A guides learners through a staged progression from station 0 to 6. Learners work the stations: they identify major minerals in rotating thin-section imagery, generate labeled datasets, train and interpretable K-nearest-neighbor classifier, segment mineral grains using SLIC phase-segmentation annotator, build knowledge base using the lunar olivine oxide data extractor, and interrogate a lunar sample agent as a web-augmented literature search and synthesis. In doing so, learners experience what AI tools can and cannot do, including verifying extractions against Apollo-era PDFs and catching errors, and watching the agent’s tool trace. The verified outputs of station work, including phase annotations, lunar olivine oxide CSVs checked against printed page, corrected markdown conversions of legacy papers, flow into a shared, versioned lunar sample corpus. Thus, learners contribute to scientific work during learning: the digitization and verification of decades of lunar sample literature. The core stations operate on local hardware without continuous internet connectivity, reducing access barriers for schools, rural communities, and resource-constrained venues. Thus, the platform offers a template for public participation in education and research at the moment AI makes it possible.

Strand B equips learners to build, train, and evaluate AI models using lunar sample texts and customized datasets drawn from the platform. Through pre-built Colab notebooks, learners progress from a character-level SampleGPT to a BPE-tokenized SampleGPT, then explore post-training through supervised fine-tuning (SFT) and reinforcement learning (RL), comparing how each stage changes model behavior and performance. Learners also train and experiment with compact architectures such as NanoChat and MiniMind, using customized lunar datasets they curate or generate through the Reading the Moon platform and curriculum. The goal is to make the full model-development pipeline visible, from data preparation and tokenization to pretraining, fine-tuning, alignment, evaluation, and model comparison, while grounding every step in lunar science.

Spiraling, Hands-on AI Literacy through Lunar Science

The curriculum uses a spiraling and hands-on design that integrates AI literacy with lunar science and lunar sample datasets. Learners repeatedly revisit the same core AI concepts, including data, representation, prediction, training, evaluation, and alignment, while working with increasingly sophisticated models and increasingly rich scientific data at increasing levels of technical depth. At each stage, learners learn by doing: exploring lunar data, curating datasets, running models, changing hyperparameters, training language models, comparing outputs, identifying failures, and contributing improved data and models back to the platform. Equipped with pre-built Colab notebooks to train language models from scratch on lunar sample text, the curriculum aims to make an otherwise invisible AI system observable, manipulatable, and understandable through experience even though they might not understand every line of code. Well-designed Colab notebooks turn model training into a laboratory so learners do not need to understand all the code to understand what the AI system is doing, what choices shape it, and why its outputs have limitations while inspecting the lunar text used for training.

Learning Outcomes

The goal of the curriculum is to develop practical, critical and creative AI literacy through hands-on engagement with lunar sample data and customized datasets drawn from the platform. By the end of the curriculum, learners will be able to:

- Make and record systematic petrographic observations and clearly distinguish observation from interpretation.
- Write evidence-based petrographic reports that support interpretations of lunar history.
- Design a Lunar petrographic exhibit through artistic expression.
- Understand how AI systems are built, trained, evaluated, and improved.
- Understand how training data, tokenization, model size, training objectives, and human preferences shape AI behavior.
- Build awareness that scientific AI depends on the quality, provenance, structure, bias, scarcity, and uncertainty of its data.
- Distinguish between AI-generated output, scientific observation, interpretation, hypothesis, and evidence.

Audience and Prerequisites

This curriculum is written for advanced secondary or lower-division undergrad students and assumes no prior geology as a compressed multi-day workshop. No prior knowledge of machine learning is assumed either.